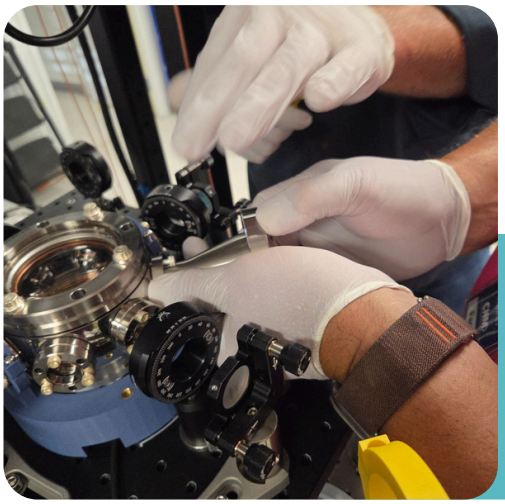
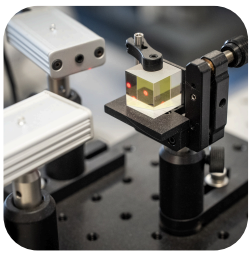
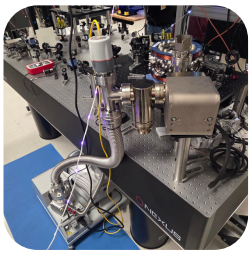
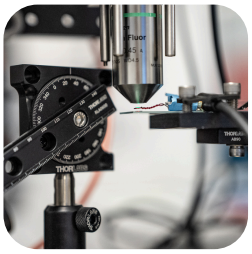


QuLL

Quantum Learning Lab Equipment Overview







Magneto-Optical Traps

- Four Rb-85 MOTs build from the Vescent Laser and the UHV chambers
- below

Lasers

- Vescent D2 780nm (108mW) Laser and Control Electronics
- Labs Electronics Dlnsec pulsed 520nm laser (down to 300ps)
- ThorLabs NPL45B 450nm diode pulsed laser (5-39ns)
- ThorLabs NPL52C 520nm diode pulsed laser (6-129ns)
- ThorLabs L405P20 405 nm, 20mW laser diode
- ThorLabs L658P040 658nm, 40mW laser diode
- Compact Laser Modules at 405nm, 532nm, 635nm, and 639nm

Optics

Four optical tables
Full suite of optical components (lens, mirrors, gratings, opto-mechanical mounts)
Polarized and non-polarized beam splitter (visible and NIR)
780nm EOM (3 GHz) and AOM (80 MHz and 200 MHz)
BBO Crystals (Type I and Type II) 405nm to 810 nm

ThorLabs Education Kits

Michelson Interferometer
Portable Optical Tweezers
Time Resolved Absorption Spectroscopy
Quantum Optics with Polarized Entanglement Expansion

Vacuum System

Three high vacuum systems (10^-7 torr) with 1 cubic foot chambers
One ultra-high vacuum 6-inch spherical-octagonal chamber (10^-10 torr) with Ion / TSP pump
Three ultra-high vacuum 4.5-inch spherical-octagonal chambers (10^-9 torr) with Ion Pumps.
Helium Leak Checking System

Sandia National Labs - Qick-Dawg diamond nv microscopy/magetometer